# Embedded and IOT Systems


E&ICT Academy, IIT Kanpur
A Joint Initiative of MeitY & IIT Kanpur

# Today's Agenda

- What is Node-RED?
- What is flow-based programming?
- History of Node-RED
- Basic nodes
- Installation of Node-RED

# Introduction about NodeRED

# Node-RED

# Everything you need to know about Node-RED

- Node-RED is a programming tool for wiring together hardware devices, APIs and online services in new and interesting ways.
- It provides a browser-based editor.
- Since early 2014, the mobile has overtaken the personal computer (desktop/laptop) as the leading device used to navigate the Net.
- Along with the mobile, a number of other portable devices that connect to the Internet have also started proliferating at a very quick rate.
- Nowadays, most of us carry or possess at least one Internet-based device and mobile. So, the Internet of Things (IoT) now doesn't only mean different 'things', but has evolved into 'intelligent things' which have onboard computation and network connections. Most importantly, they have the capability to sense the environment around us and, accordingly, act intelligently. These devices are now referred to as connected devices, smart objects or the Web of Things.

# Everything you need to know about Node-RED

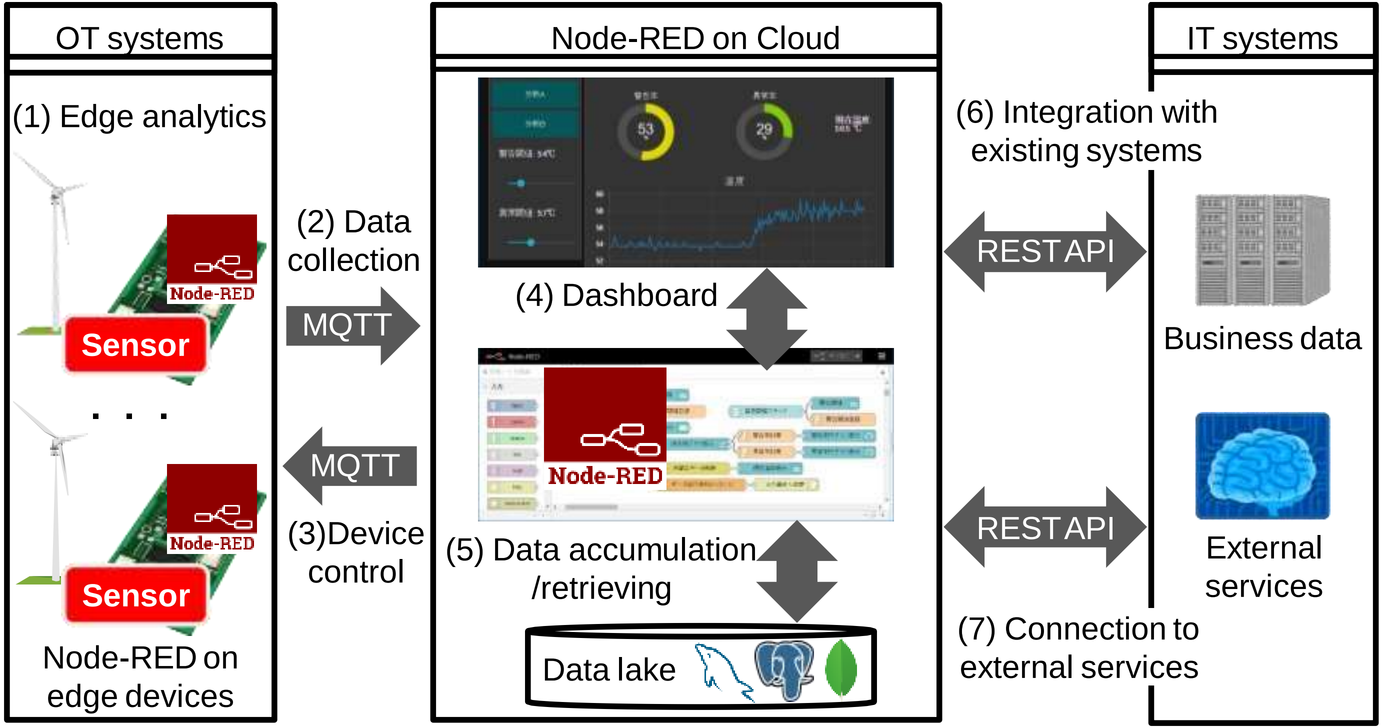
- This adaptation of technology has led to many developments in the visual programming and visual coding arenas.
- As a result, it has evolved into a visual programming/coding tool based on the already popular Node.js (a server side Java Scripting platform), mainly targeting the Internet of Things space.

# What is Node-RED?

- Flow based Programming which realizes quick development.
- Node-RED is a programming tool for wiring together hardware devices, APIs and online services in new and interesting ways.
- It provides a browser-based editor that makes it easy to wire together flows using the wide range of nodes in the palette that can be deployed to its runtime in a single-click.
- Various connectors(Node-RED nodes) to add functionalities
- Open source software under Linux foundation

# What does Node-RED realize?

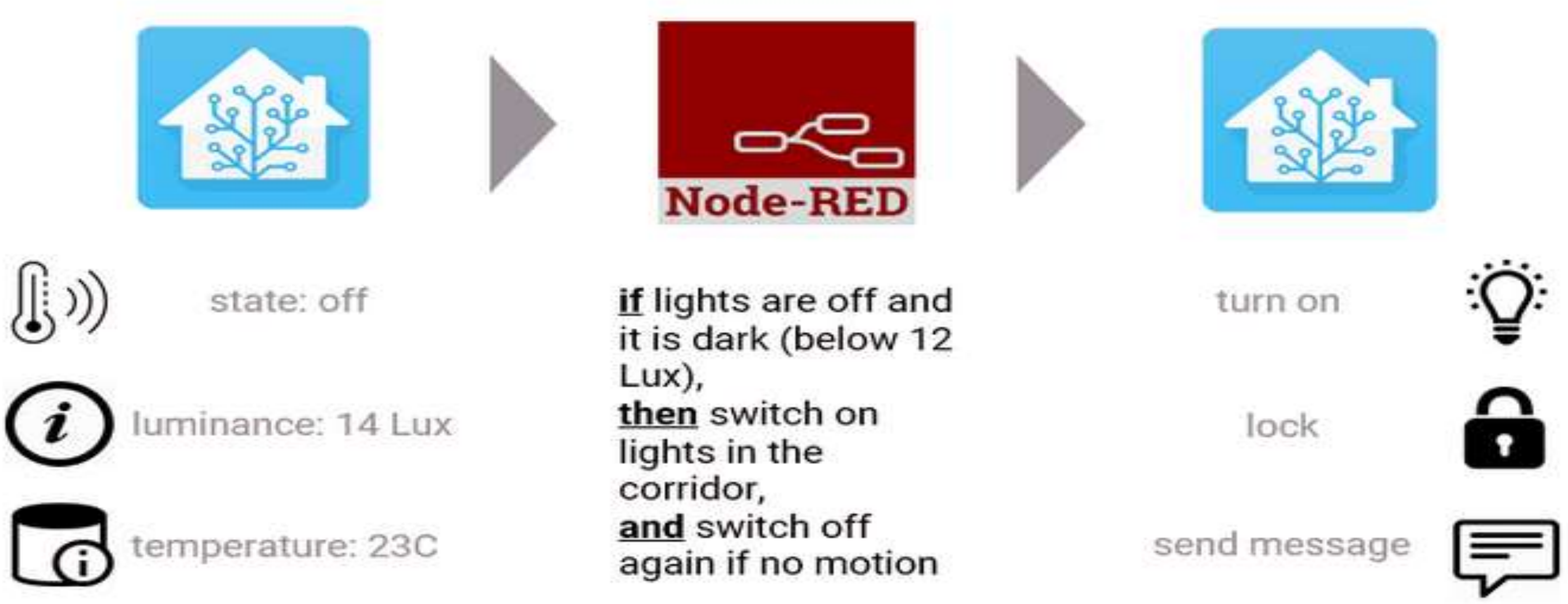
## Node-RED has key features to realize IoT system.



# What is flow-based programming?

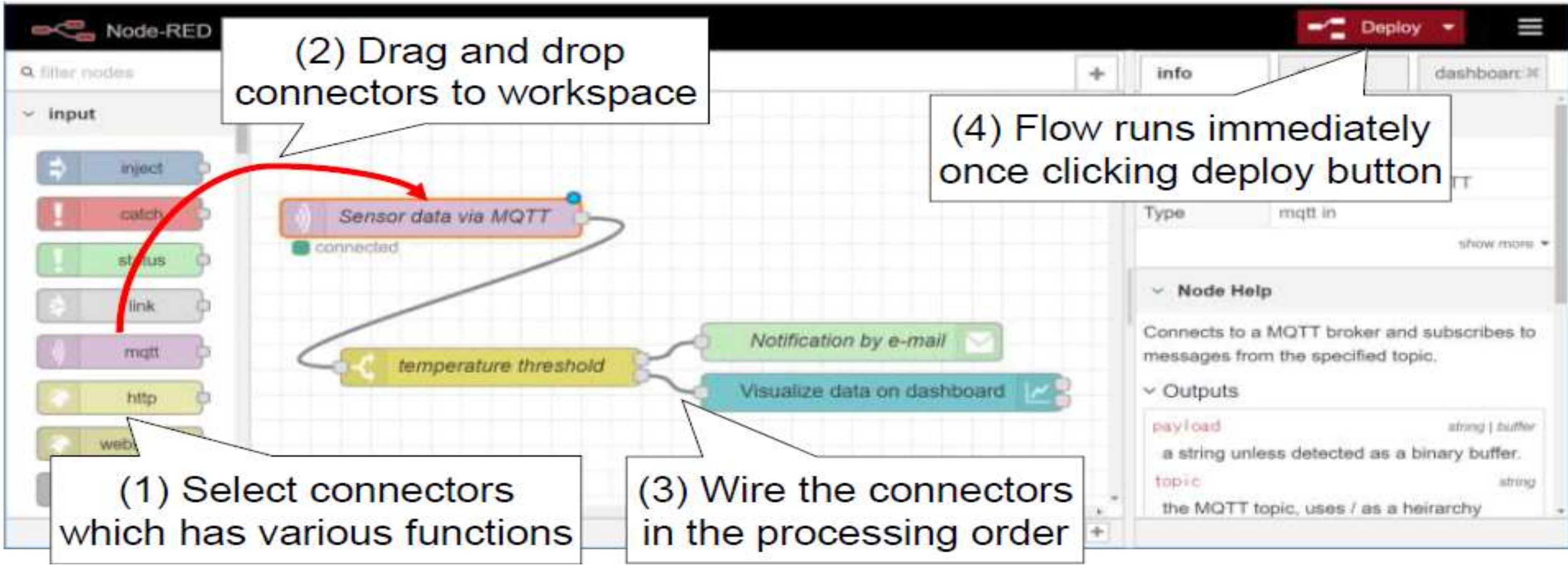
- **Flow-Based Programming (FBP) is a slightly different approach to thinking about programming.**
- **The basic premise is to decompose the problem up into several components: data, processes, and the network.**
- **In FBP the applications are the collections processes, which linked using data Information Packets that travel between processes though defined connections.**
- **Compared to the usual way of thinking about the coding, flow-based programming is much more visual and you can almost draw the code.**
- **This makes it easy to learn and especially great for applications that need more mass adoption.**

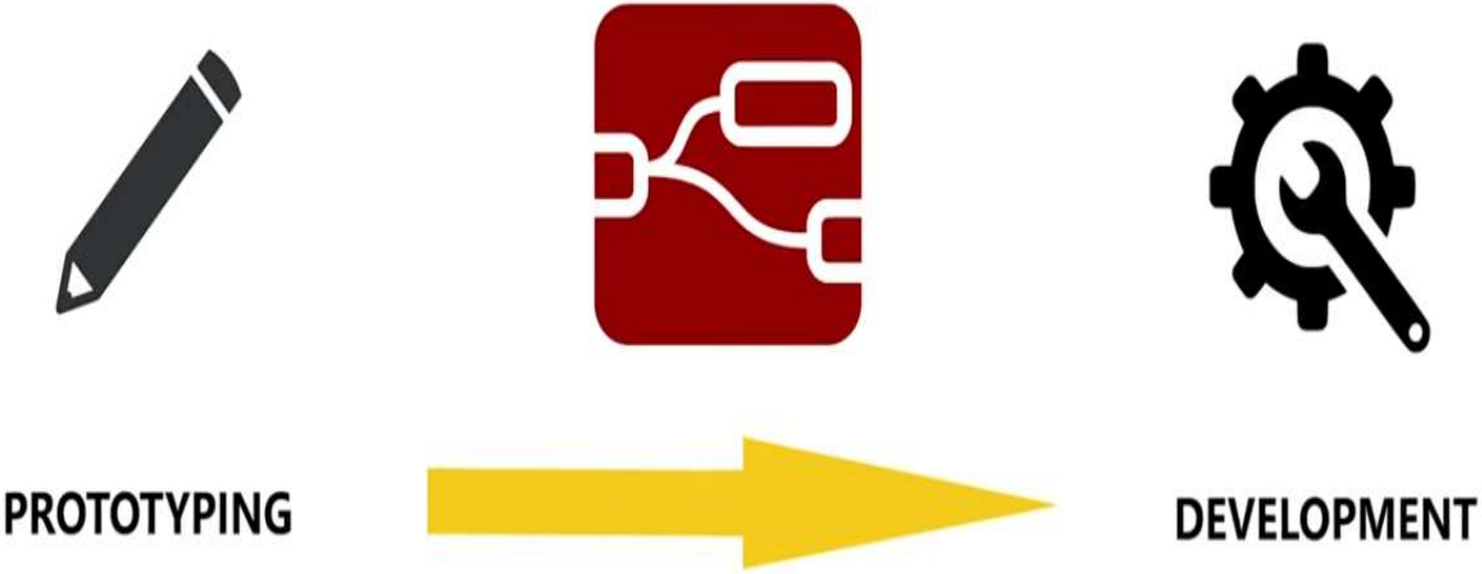
# Architecture of interaction between Node-RED and Home Assistant



# What is Node-RED?

## Visual programming tools for IOT Applications

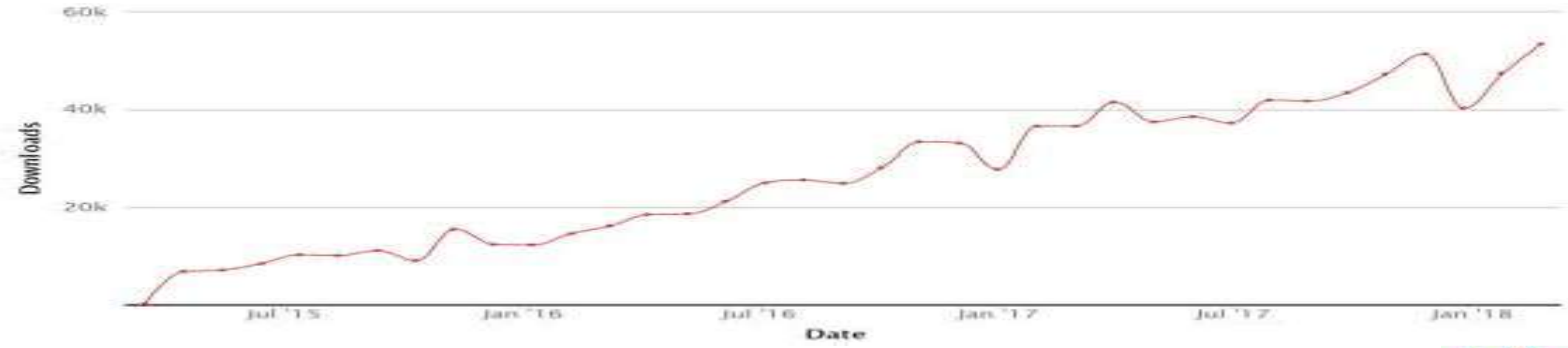



PROTOTYPING
DEVELOPMENT

# History of Node-RED

**Node-RED has been more open and popular as open source software for both edge and cloud environments.**

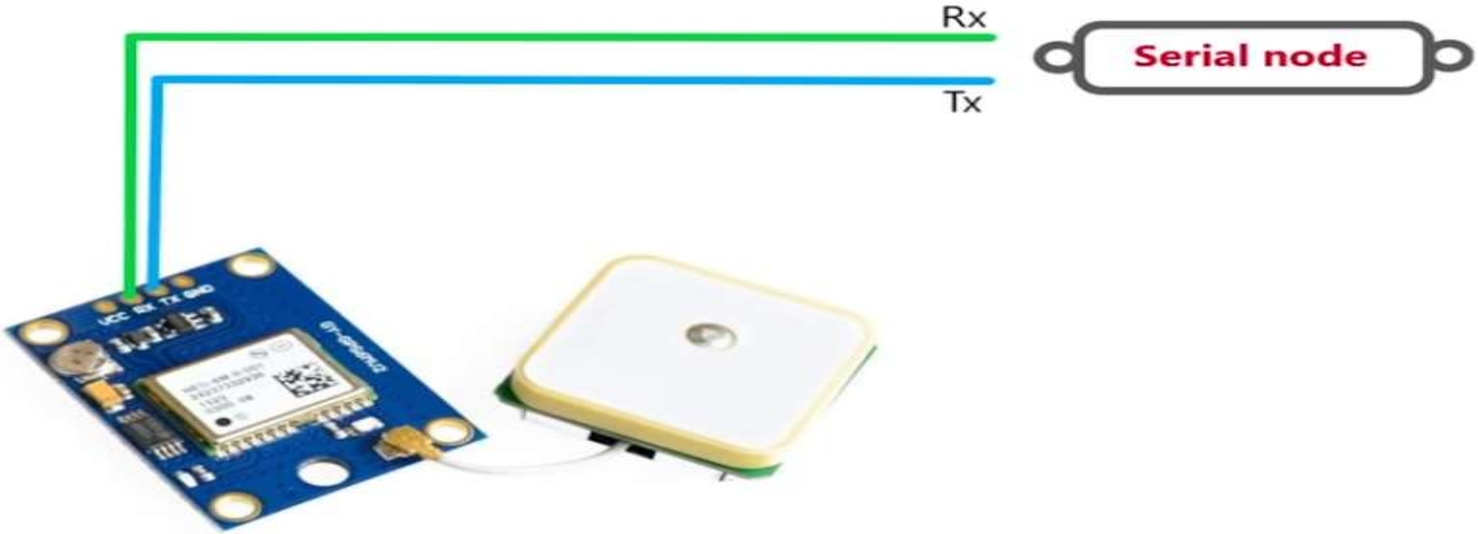
- **March 2014: Released in QCon London.**
- **June 2014: Included in the catalog on IBM Cloud.**
- **November 2015: Pre-installed in Raspberry Pi.**
- **October 2016: Moved to Linux Foundation.**
- **As of March 2018: Downloaded 50,000 times a month.**



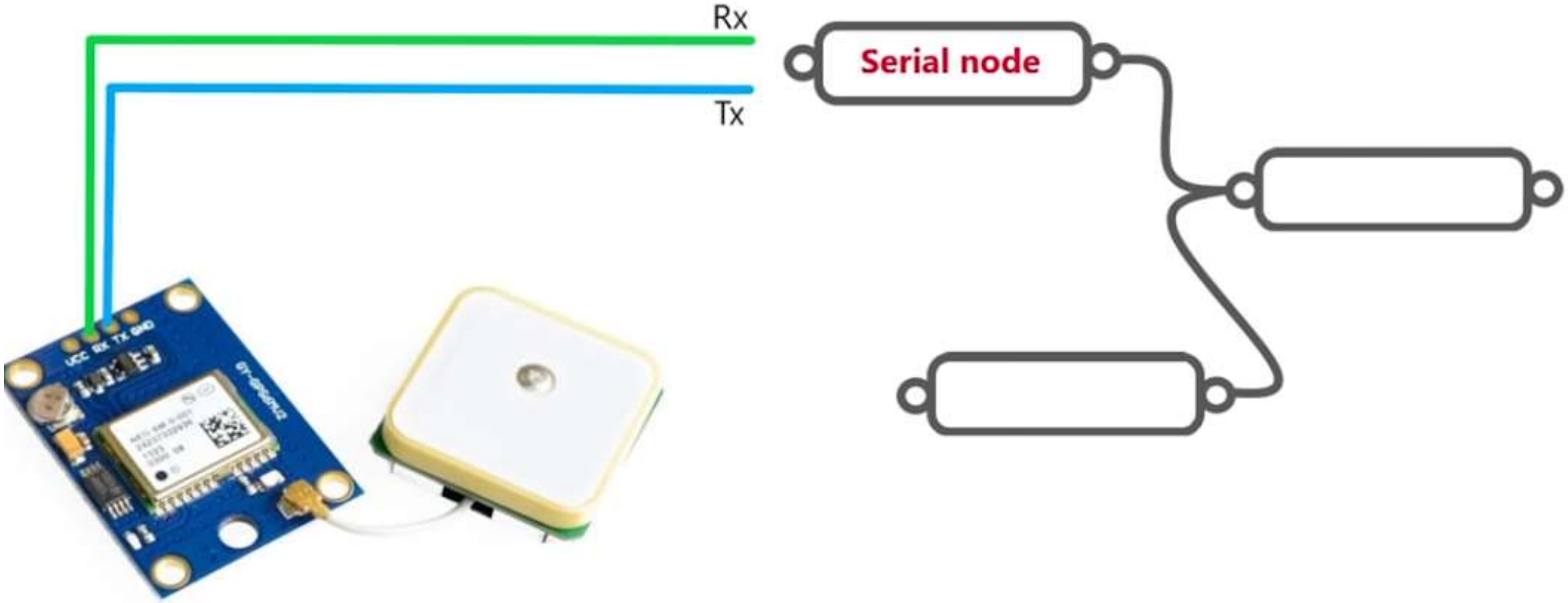

**The number of download from npm repository**

# EXAMPLE

# EXAMPLE



# EXAMPLE



# Why Node-RED?

**(1) Rapid development for IoT applications**

- IT and OT engineers can easily create IoT application within couple of hours without coding.
- Same development styles in both edge and cloud environments


**(2) Standard technologies in industrial IoT**

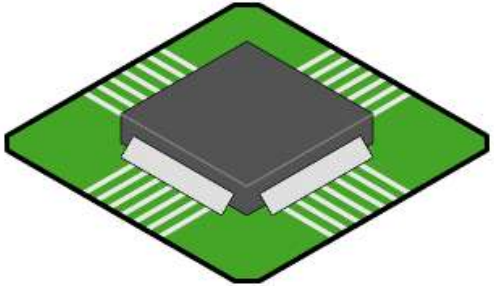
- Node-RED supports essential IoT technologies. (e.g. REST API, WebSocket and MQTT)
- Pre-installed software on standard edge devices

**(3) Open community**

- 73 contributors including IBM, Sense Technic, and Hitachi are developing Node-RED on GitHub.
- 3rd party 1,345 connectors

# Who can host Node-RED?

**Run Locally**



**On a device**

- Raspberry Pi
- BeagleBone Black
- Interacting with Arduino
- Android

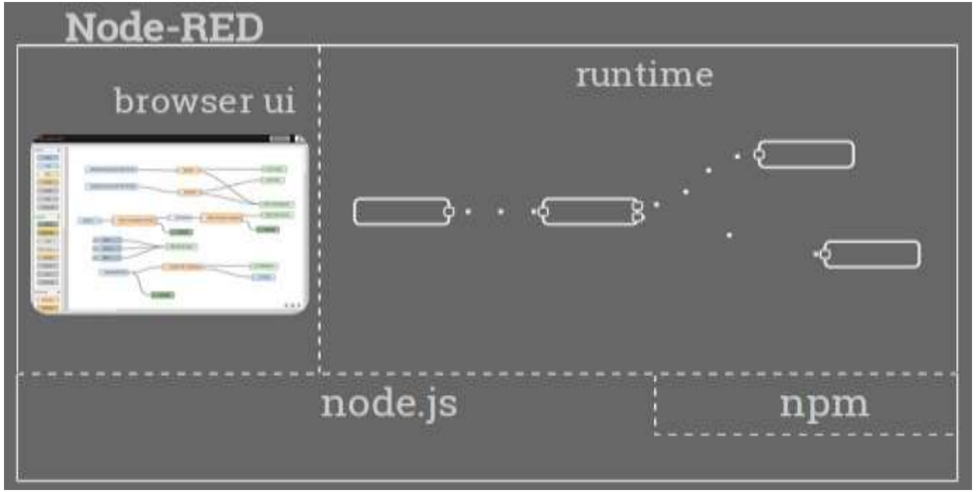
**In the cloud**

- IBM Cloud
- Amazon Web Services
- Microsoft Azure

# Features of NodeRED

# Architecture of Node-RED

- **Node.js v8-engine driven;** so it's fast
- **Event-driven, asynchronous io;** it's all about the events
- **Single-threaded event-queue;** built for fairness
- **Javascript front and back;** only one language runtime to deal with.



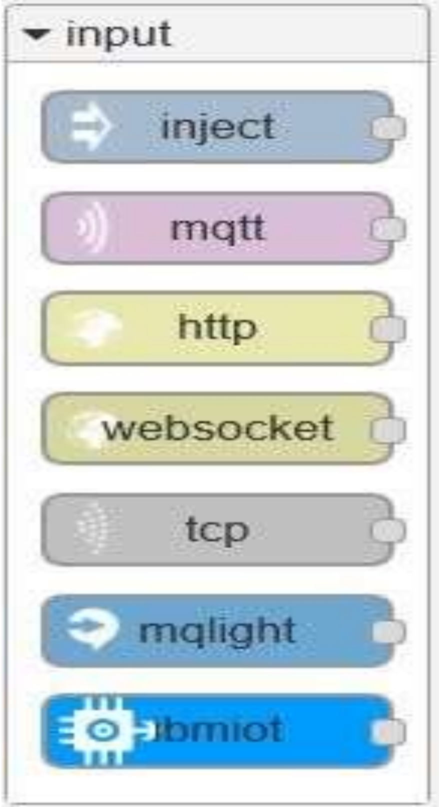
# Basic Node types


- Inject node
  - Allows manual triggering of flows
  - Can inject events at scheduled intervals


- Debug node
  - Show message content; either payload or  entire object

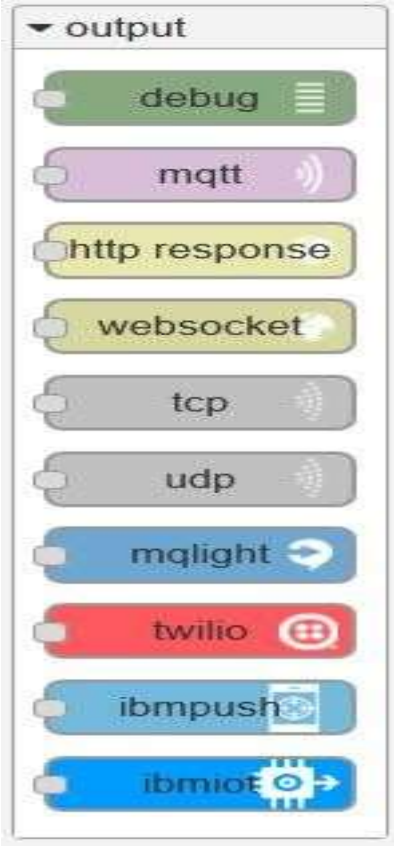

- Template/ Function Node
  - Modifies the output based on a Mustache  Template

# Input Nodes



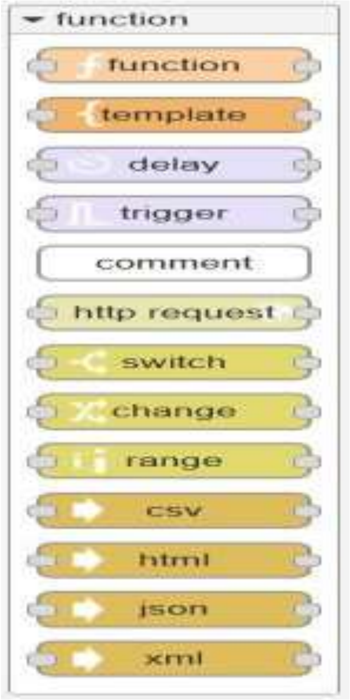

- HTTP – Act as an HTTP endpoint; great for building RESTful services
- IBMIOT – Receive messages from an attached IOT Foundation account
- Also can receive from Websockets, MQTT (pick your own broker), TCP and MQ Light

# Output Nodes



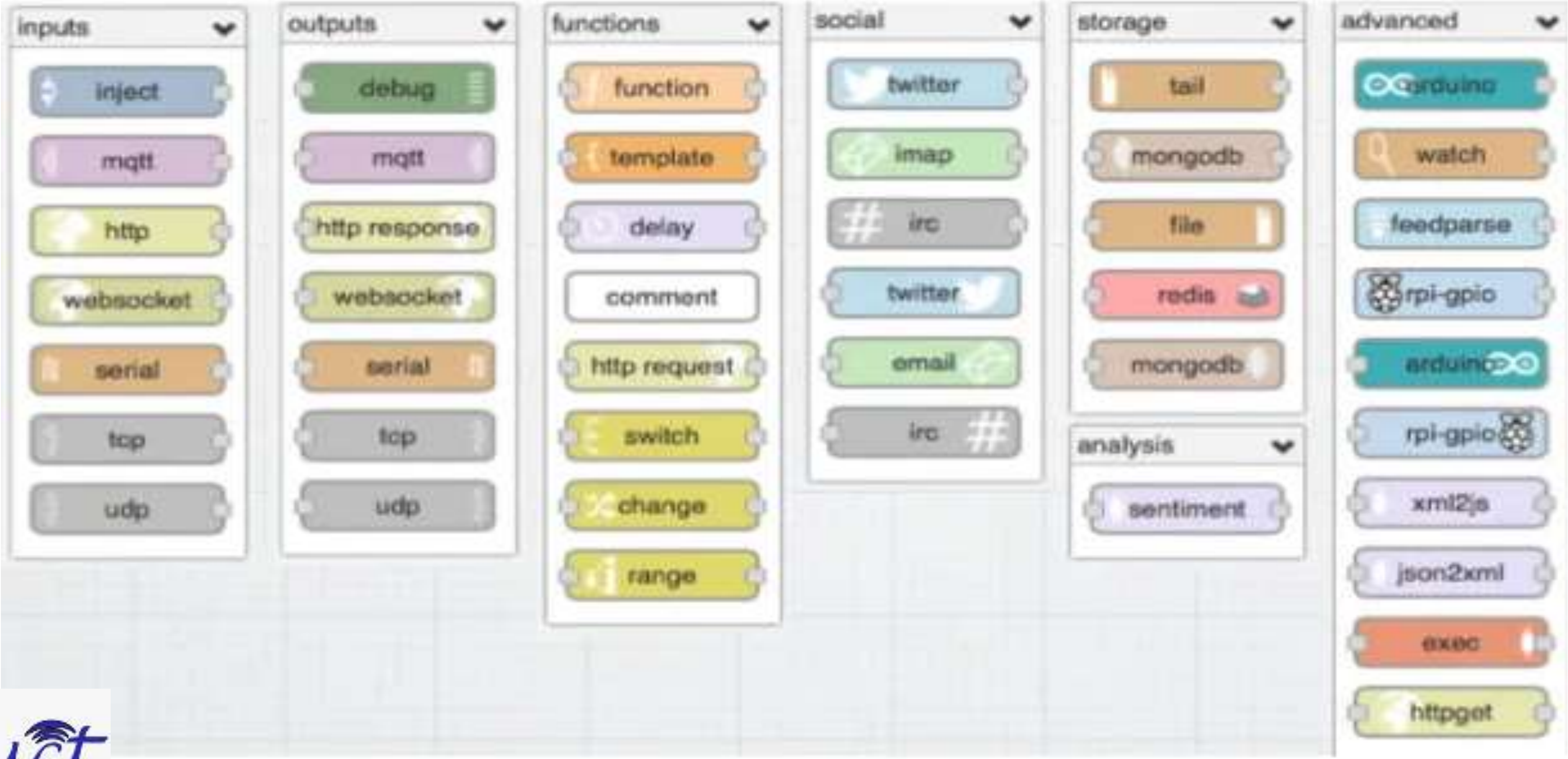

- **HTTP Response; required as the final node when the input comes from an HTTP Request**
- **IBMIOT – send events out to the attached IOT Foundation account**
- **Twilio – send SMS messages via the Twilio service**
- **IBM Push – Send Push notifications to mobile devices**
- **Also can send requests through TCP, UDP, MQLight, WebSockets.**

# Function Node Types



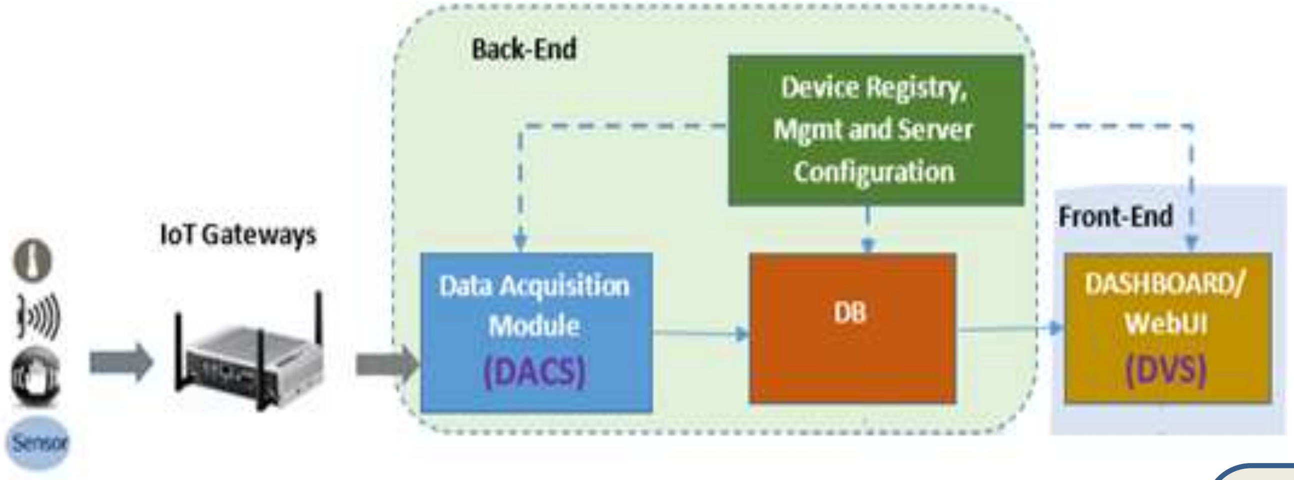

- **Function node**
  - **Run user-defined node.js code on the messages going by**
  - **Uses vm.createScript under the covers to sandbox execution**
  - **Console, util, Buffer included for convenience**
- **Switch**
  - **Change flow to different options based on a comparison**

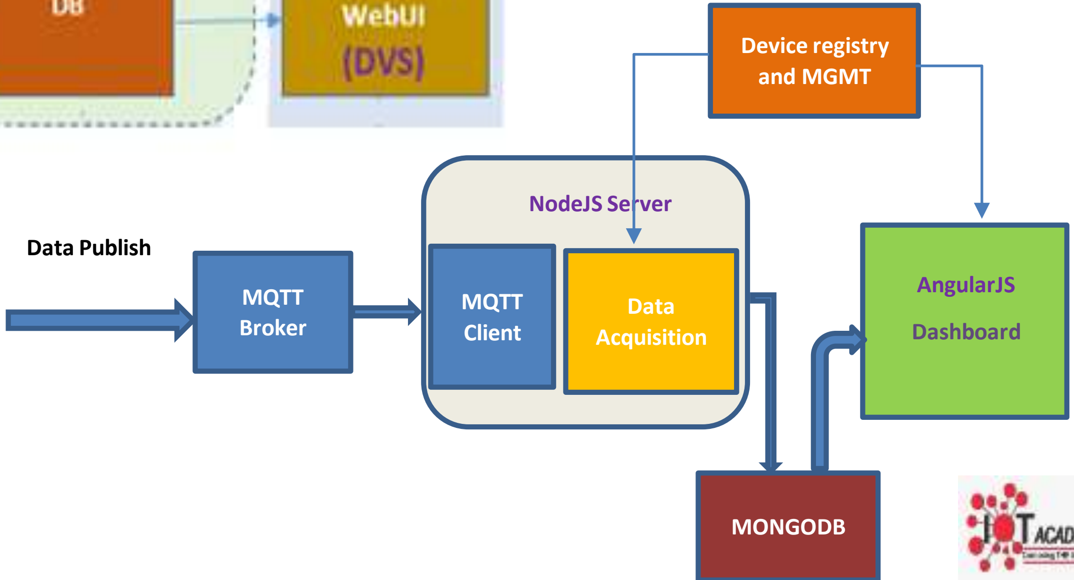
# List of Nodes



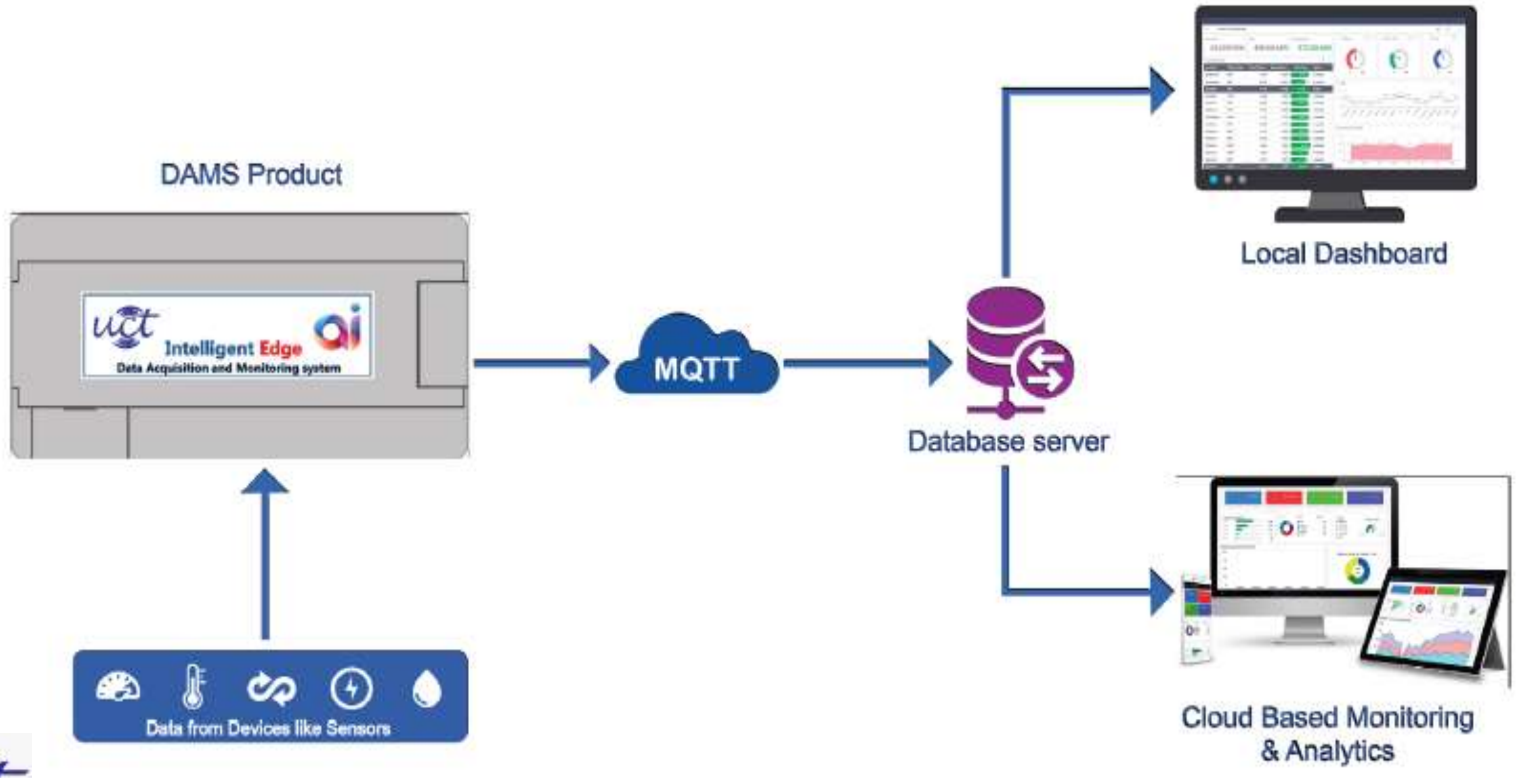
# Need of a tool : NodeRED

# IOT DATA FLOW IN ANY SOLUTION

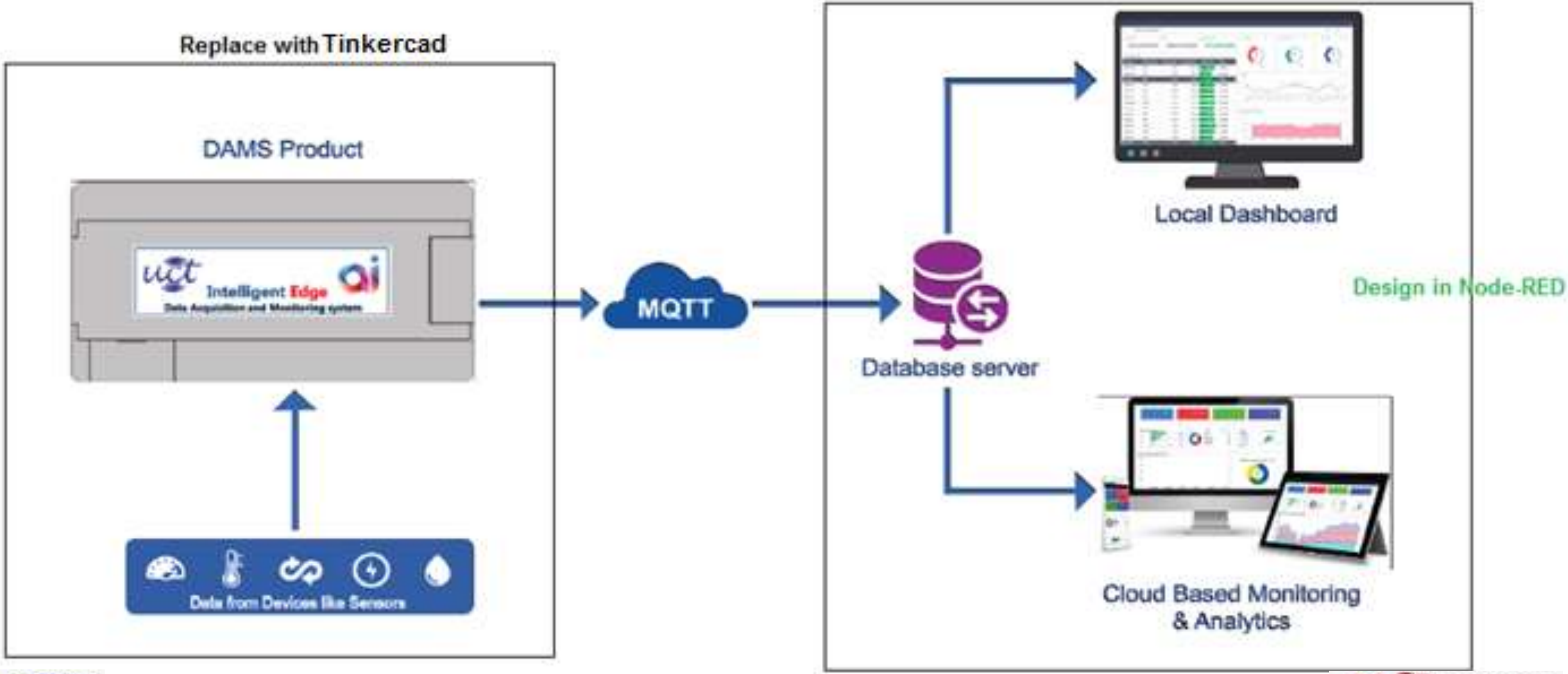


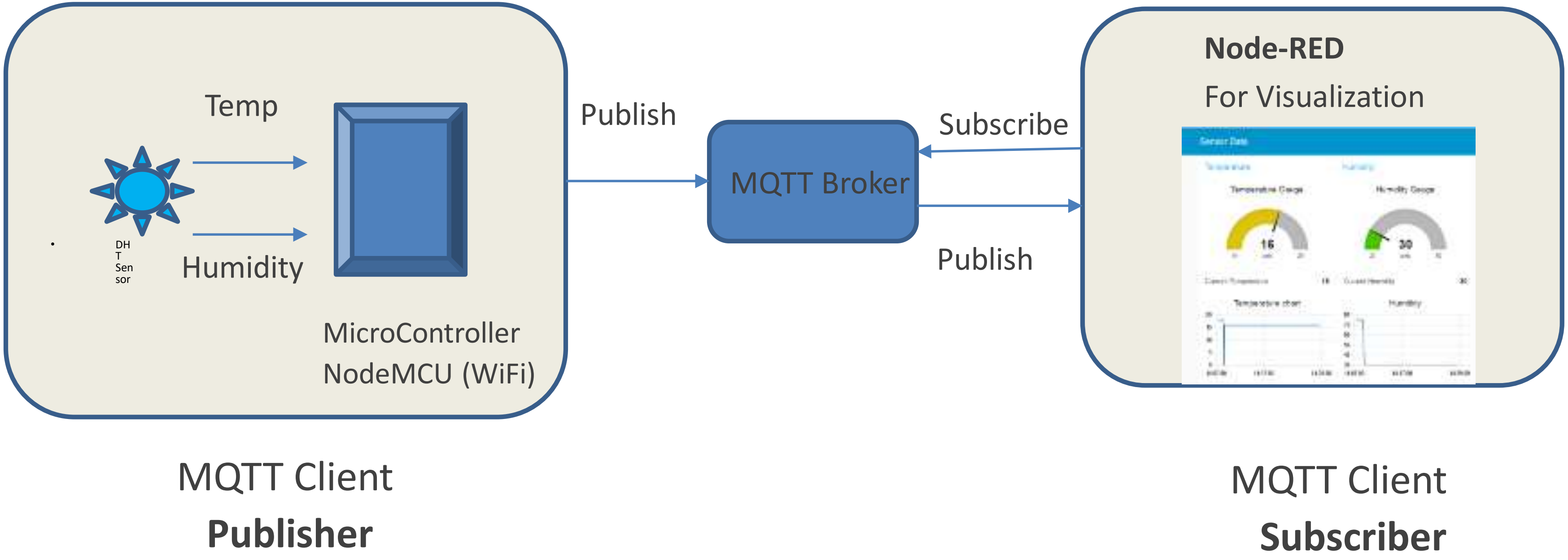
# Generic IOT Monitoring Solution



# Approach to replace HWs with SW



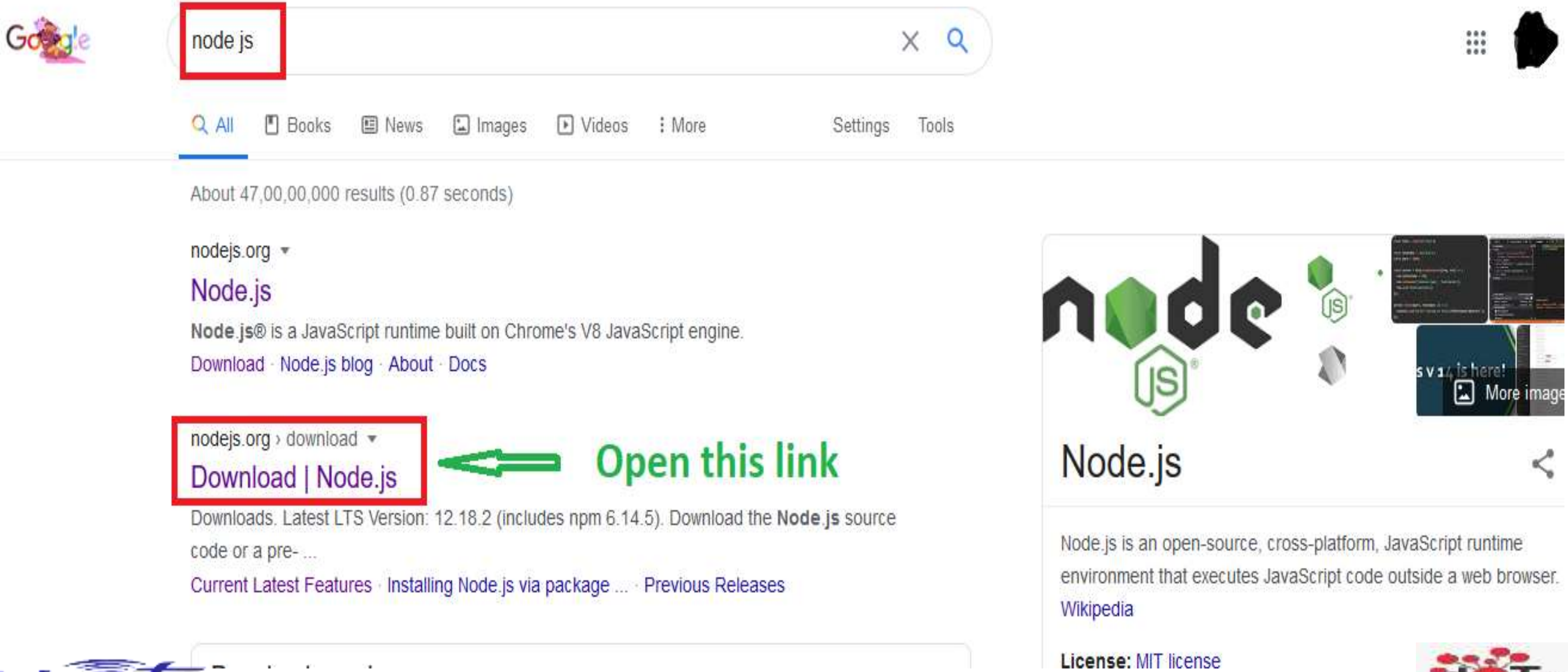
# Temperature Humidity Monitoring



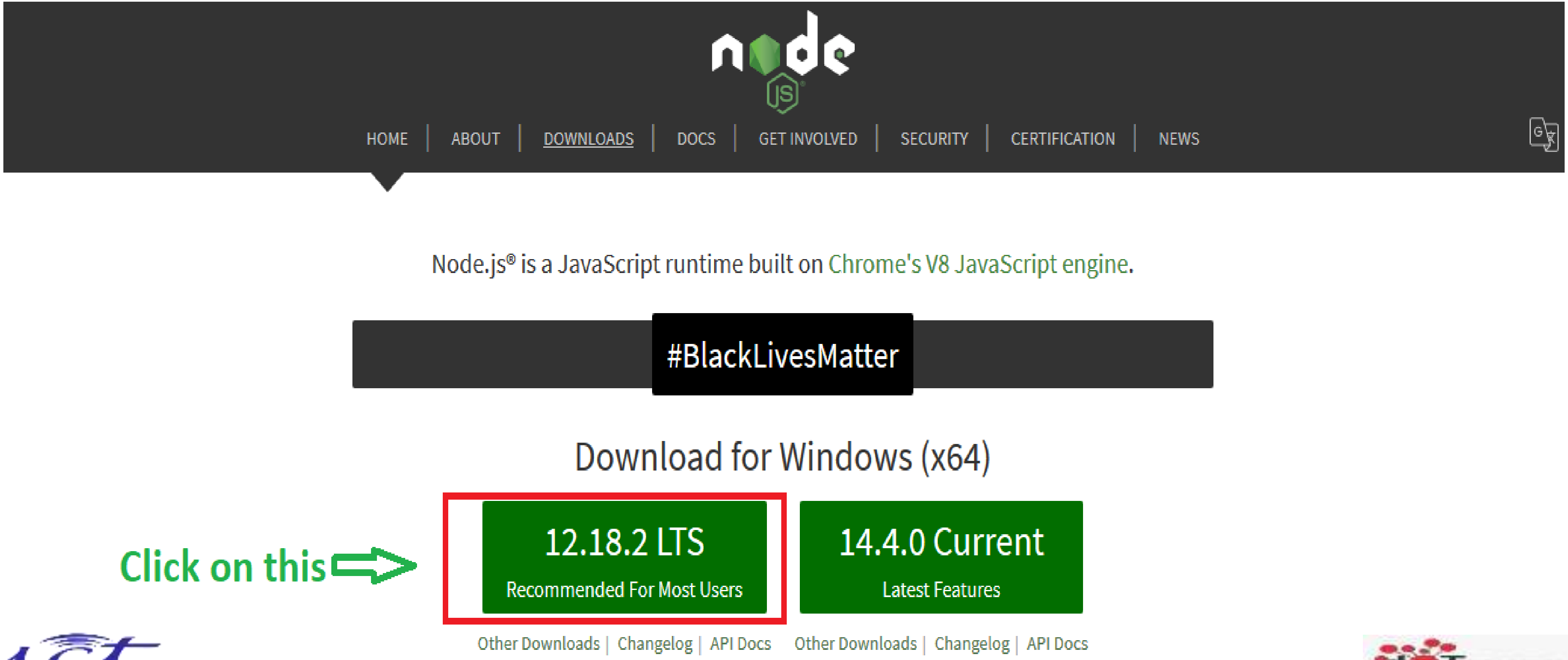
# Installation of NodeRED

# Node-RED Installation Steps for Windows

## Step 1. Search node js in Google and open downloads



# Step 2.
# (a) For Windows

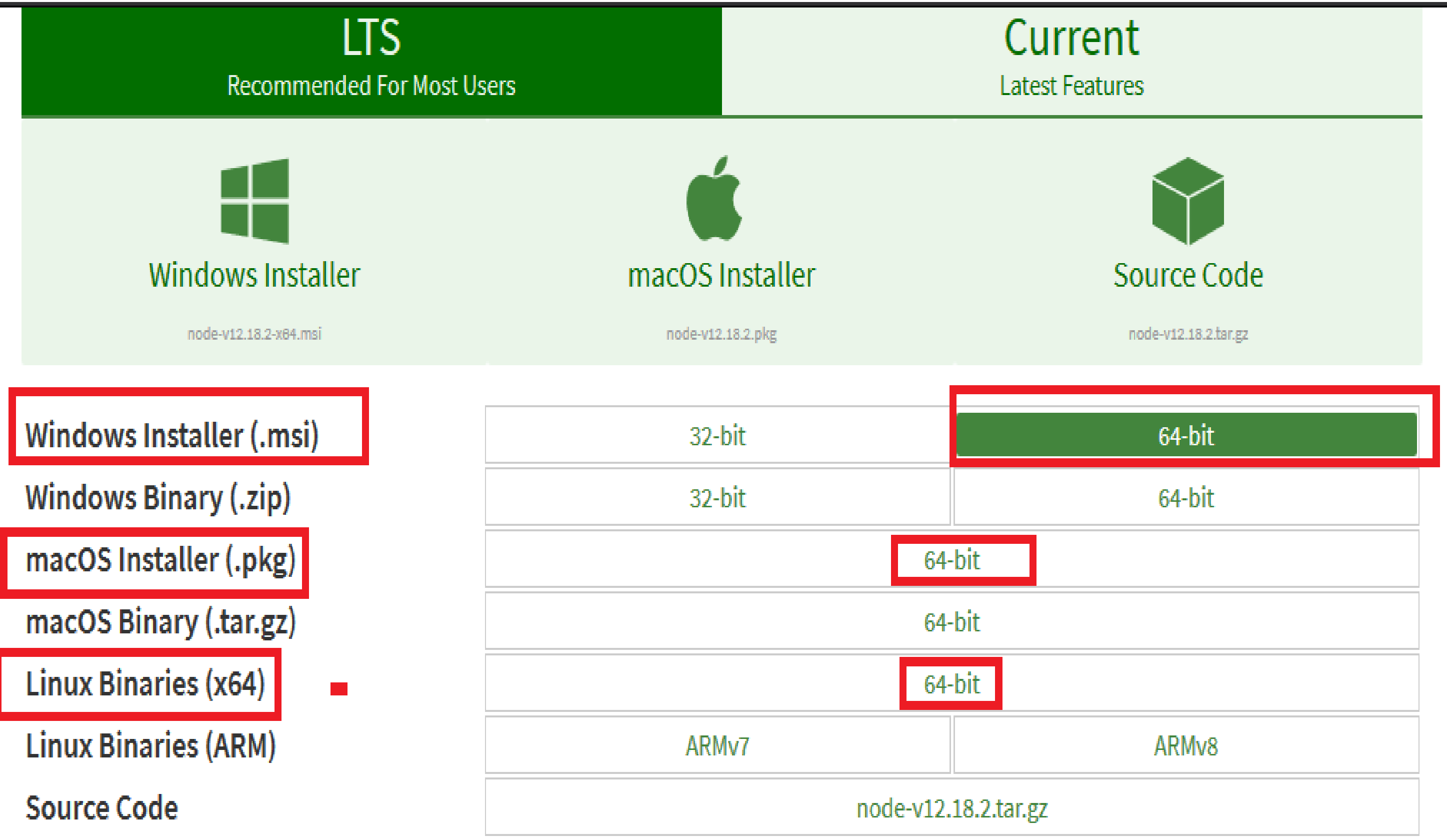



Click on this

# Step 2.
# (b) For Linux

# Step 2.
# (c) For Mac

# Step 3. Select Installer

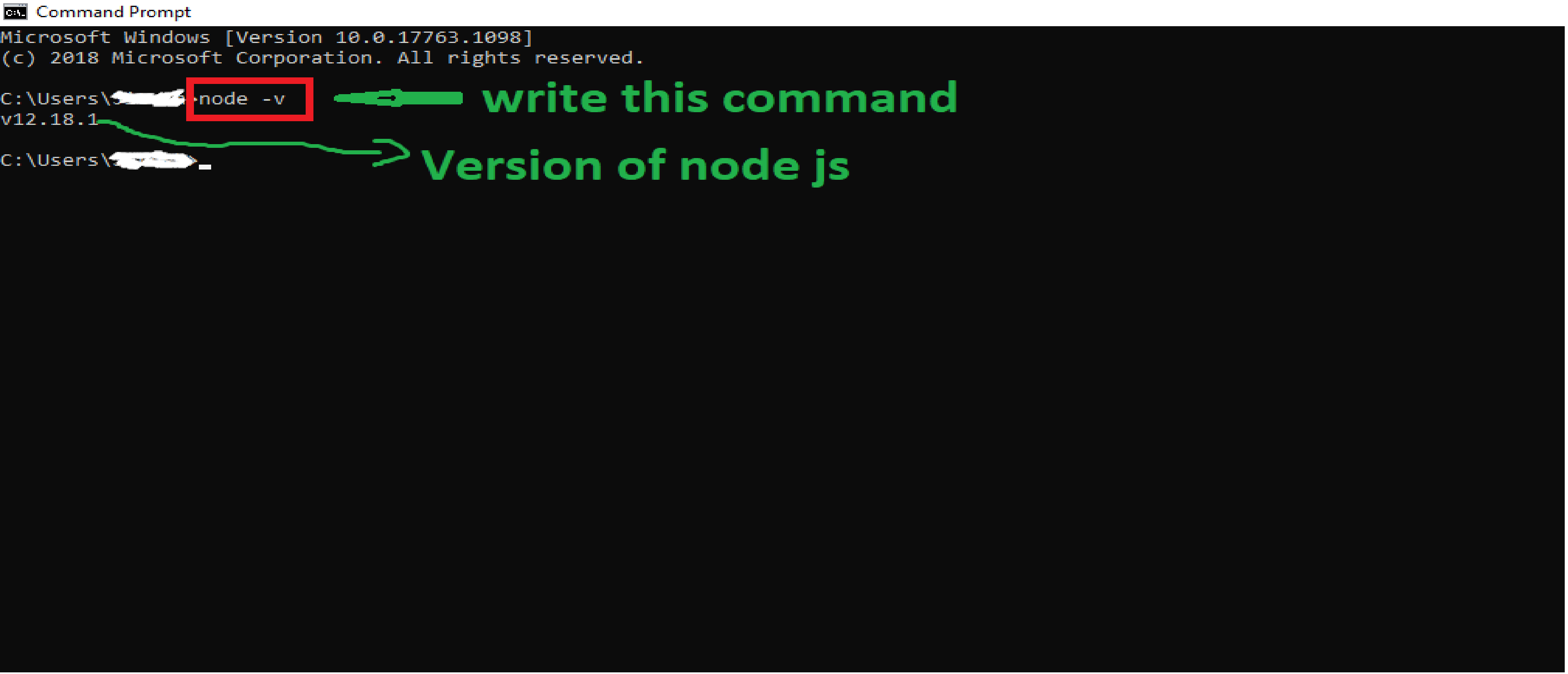


# Step 4. Open Command Prompt

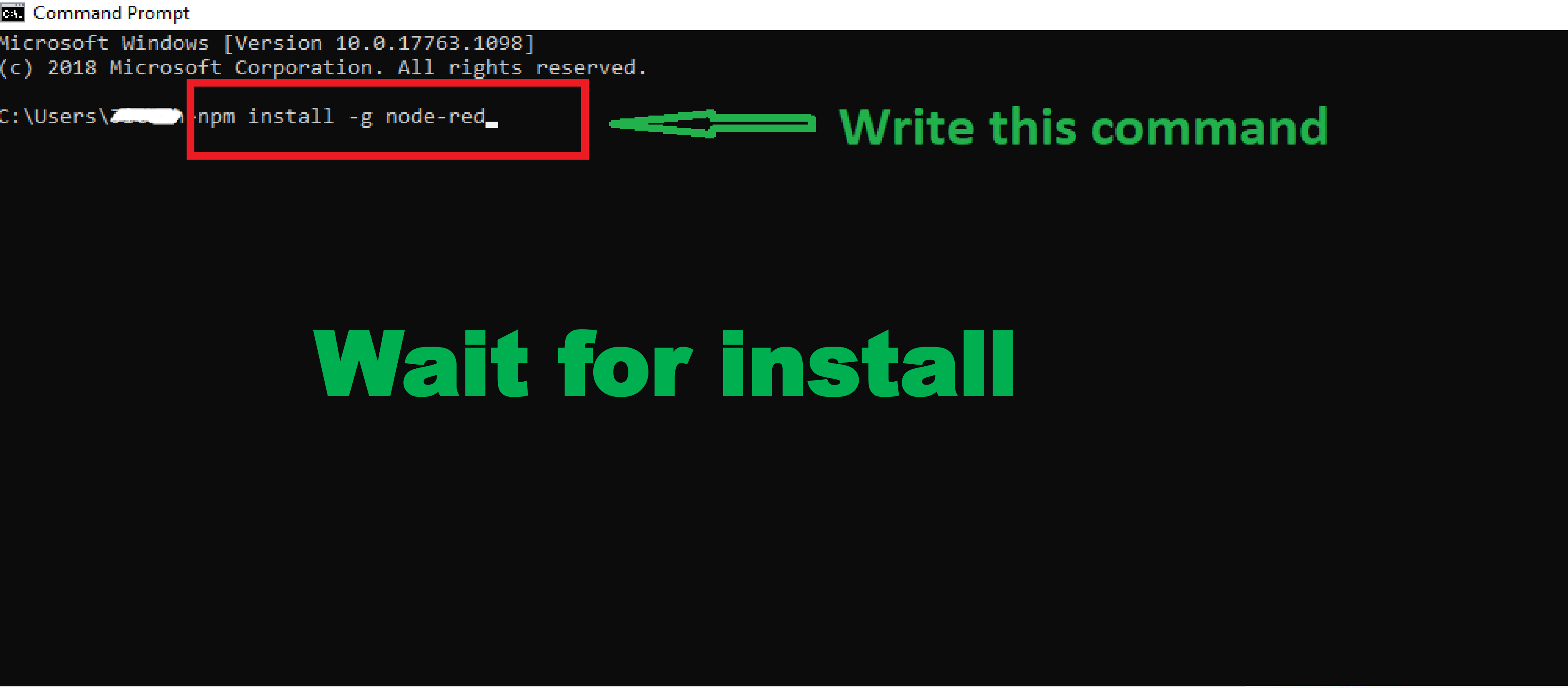
Command Prompt

```
Microsoft Windows [Version 10.0.17763.1098]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\[redacted]>
```

# Step 5. write node -v command and check version of node js



# Step 6. write command npm install –g node-red



# Step 7.write node-red

Command Prompt - node-red

```
Microsoft Windows [Version 10.0.17763.1098]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\[redacted]>node-red
```

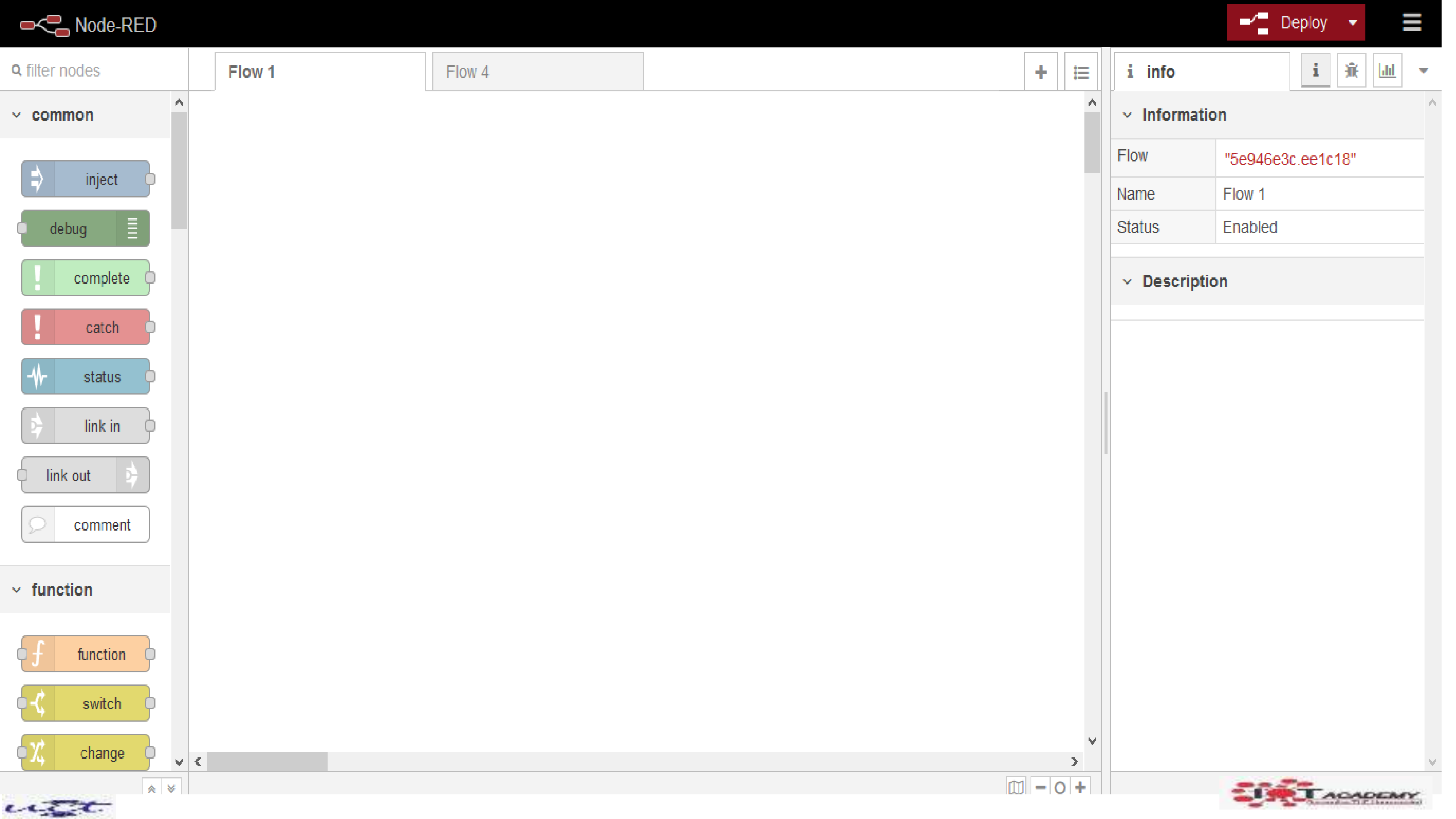
Wait for some times

# Step 8. Wait for this window and copy link at Browser

Select node-red

```
Microsoft Windows [Version 10.0.17763.1098]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\[redacted]>node-red
30 Jun 20:31:47 - [info]

Welcome to Node-RED
===================

30 Jun 20:31:47 - [info] Node-RED version: v1.0.6
30 Jun 20:31:47 - [info] Node.js  version: v12.18.1
30 Jun 20:31:47 - [info] Windows_NT 10.0.17763 x64 LE
30 Jun 20:31:51 - [info] Loading palette nodes
30 Jun 20:32:07 - [info] Dashboard version 2.15.3 started at /ui
30 Jun 20:32:07 - [info] Settings file  : \Users\[redacted]\.node-red\settings.js
30 Jun 20:32:07 - [info] Context store  : 'default' [module=memory]
30 Jun 20:32:07 - [info] User directory : \Users\[redacted]\.node-red
30 Jun 20:32:07 - [warn] Projects disabled : editorTheme.projects.enabled=false
30 Jun 20:32:07 - [info] Flows file     : \Users\[redacted]\.node-red\flows_jitesh.json
30 Jun 20:32:07 - [warn]

---------------------------------------------------------------------
Your flow credentials file is encrypted using a system-generated key.

If the system-generated key is lost for any reason, your credentials
file will not be recoverable, you will have to delete it and re-enter
your credentials.

You should set your own key using the 'credentialSecret' option in
your settings file. Node-RED will then re-encrypt your credentials
file using your chosen key the next time you deploy a change.
---------------------------------------------------------------------

30 Jun 20:32:07 - [info] Starting flows
30 Jun 20:32:07 - [info] Started flows
30 Jun 20:32:07 - [info] Server now running at http://127.0.0.1:1880/
30 Jun 20:32:08 - [info] [mqtt-broker:1c35b1b.3a202ce] Connected to broker: mqtt://test.mosquitto.org/:1883
```

Copy this link in Browser


Node-RED
Deploy
filter nodes
Flow 1
Flow 4
info
common
inject
debug
complete
catch
status
link in
link out
comment
function
function
switch
change
Information
Flow
"5e946e3c.ee1c18"
Name
Flow 1
Status
Enabled
Description

# Hands-ON

# Node-RED Hello World



- **When you click on the Inject Node, it sends and event through the flow – triggering the template node and sending the result to the Debug node**

# Other Hands-ON

- Please check all exercises at
- https://cookbook.nodered.org/basic/

- Convert timestamp to array hours, minutes, day, month, year
- https://flows.nodered.org/flow/ba071f83d91d2b065074

- Export the flows copied from above exercises to your NodeRED browser
-

# Study

https://nodered.org/docs/user-guide/writing-functions

https://nodered.org/docs/user-guide/nodes